

Universität
Basel

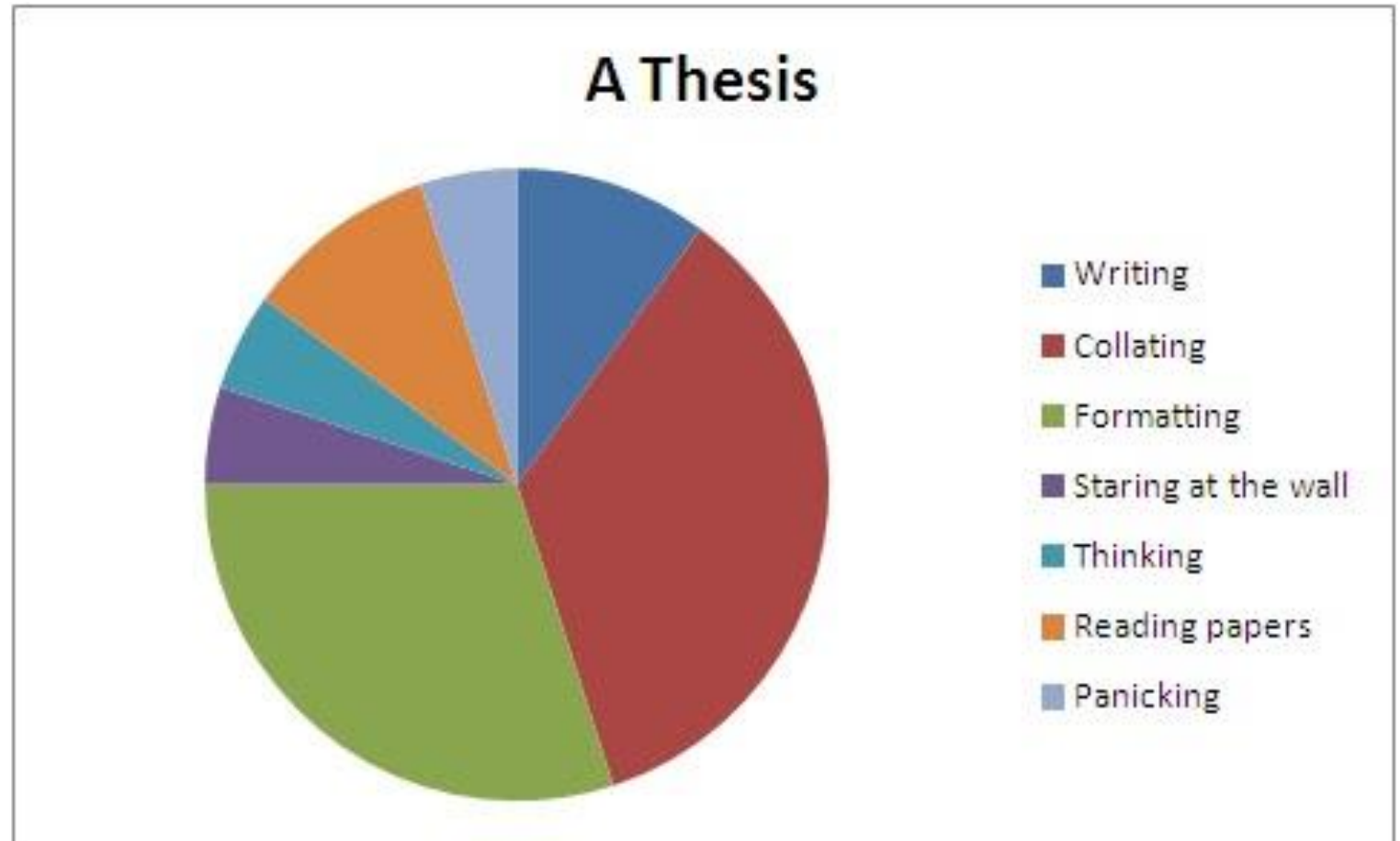
Swiss TPH



Swiss Tropical and Public Health Institute
Schweizerisches Tropen- und Public Health-Institut
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Lesson 7: Writing the Discussion and Conclusion sections



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Scientific writing Swiss TPH Riana Paola

Objectives Lesson 7

1. Examine the Discussion and Conclusion sections of research articles or theses from your field.
2. Explore the contents and structure of those sections.
3. Learn about hedging techniques to soften and strengthen claims.
4. Focus on the use of relative clauses and commas



Task: Discussion and Conclusion

Examine some articles in your field:

1. Are there Discussion and Conclusions sections? Are they separate from each other or combined into one?
2. Is there another section / are there other sections that interpret the results of the study? What are they called?
3. What does this section/do these sections contain?
4. How long is this section /are these sections?
5. How do the Results and the Discussion sections differ?

The Discussion is a section that

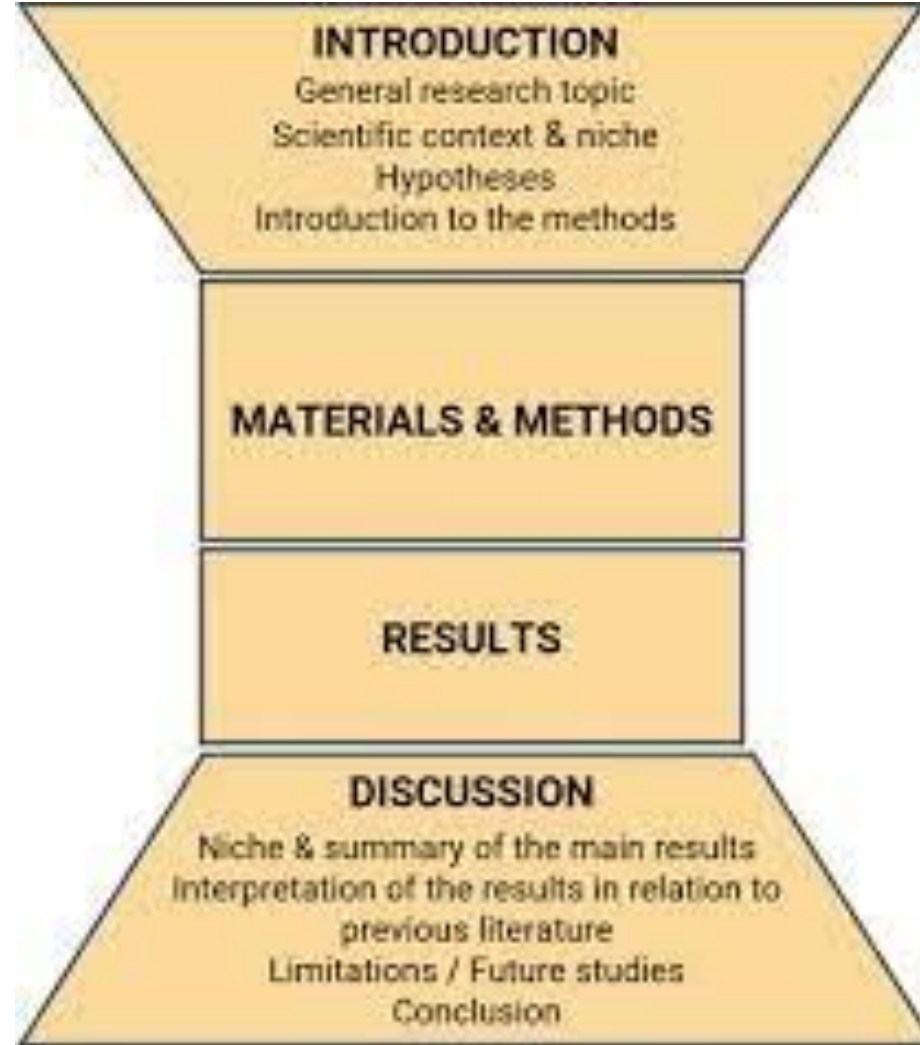
- moves from solid **facts (the results)** to **speculation (your interpretation)**
- is likely the most entertaining to read and the hardest to write.

Discussion

What do you expect from the Discussion



Conclusion section mirrors the Introduction section to establish overall text cohesion (see next slide)



Link Introduction with Discussion & Conclusion

For **overall text cohesion** and coherence, it is important that the **Introduction** and **Discussion/Conclusion are clearly linked to the Introduction section** (just like Methods and Results need to be linked).

The Introduction poses certain questions and aims, as well as indicates a gap/problem in the research (mostly).

The Discussion/Conclusions section(s) should answer these questions (and/or fill the research gap) and show how your study addresses these initial questions by saying what the findings indicate about the answers.

Establishing a clear link gives a satisfying end to the research narrative.

Introduction: Reason WHY?

Moves from general to specific



Discussion: The 'SO WHAT!'

Moves from specific to general



(Busse & August 2020)

Structure of the Discussion

This is the “**so what?**” section of your manuscript.
In the Discussion, you place your findings in the broader research context.

- **Move 1: Summarize** your (main) findings
- **Move 2: Contextualize** your study
- **Move 3: State limitation(s)**
- **Move 4: State the implications and importance** of your findings and/or make **recommendations**. This move contains the **final message sentence**, stating the **main findings and implications** in one sentence.





Purpose of Discussion section

Discussion **interprets** the data and **explains** why the **results are meaningful** in relation to **previous, related work** and the **hypothesis/research question** that was explored.

(Swales and Feak, 2012:309).

How does one interpret data?

- Identify correlations, patterns, and relationships among the data
- Discuss whether the results met your expectations or supported your hypothesis or answered your research questions.
- Contextualize your findings within previous research and theory
- Explain unexpected results and evaluating their significance
- Consider giving possible alternative explanations and making an argument for your position
- ❖ You can organize your discussion around key themes, hypotheses, or research questions, following the same structure as your results section.
- ❖ Alternatively, you can also begin by highlighting the most significant or unexpected results.

Features of the Results and Discussion

Results	Discussion
Objective	Subjective
Facts	Interpretation
Declarative statements	Speculative statements
Figures 'speak for themselves'	Your own opinion, some first-person pronouns
Past tense to refer to findings	Present tense

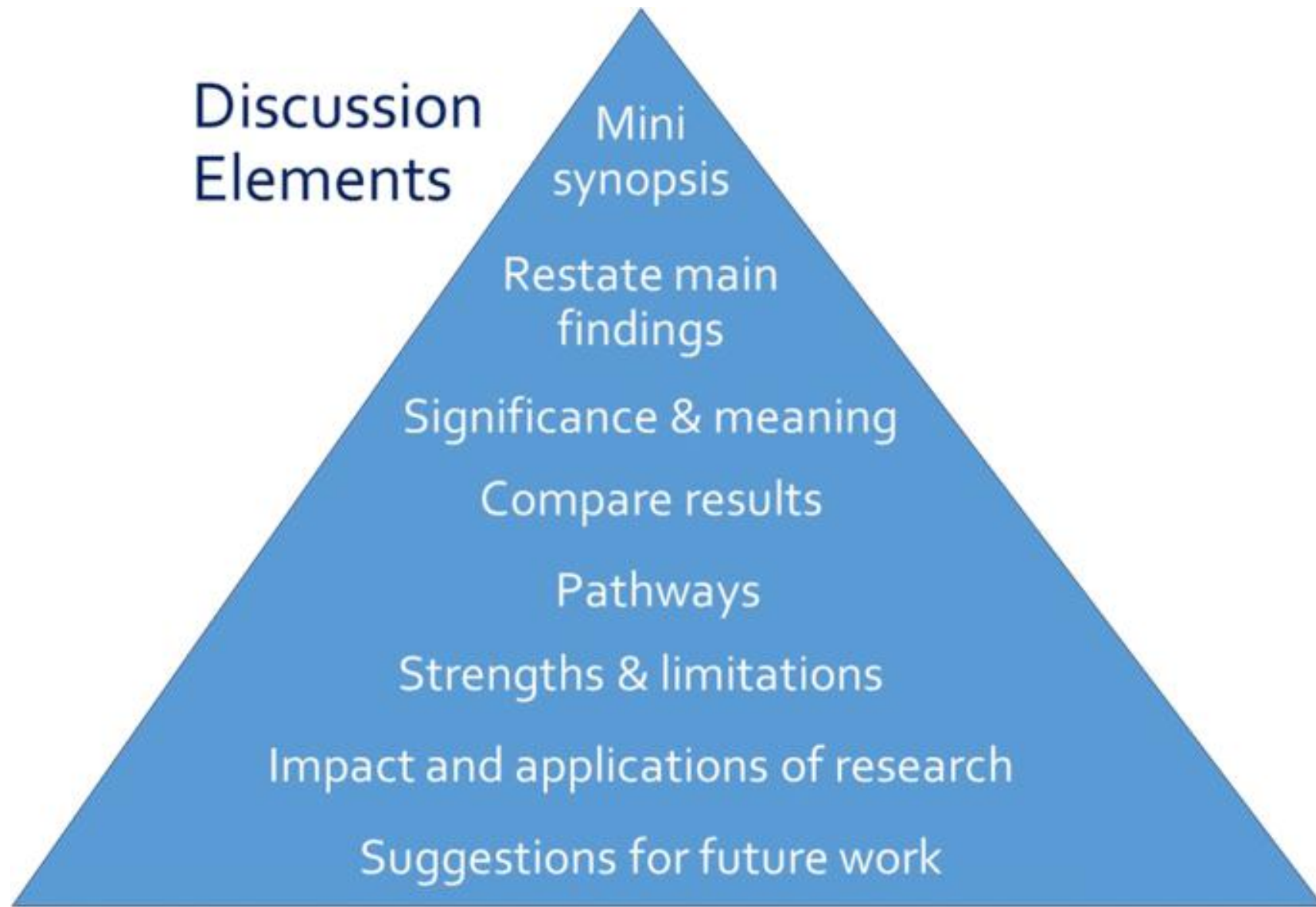


Tense use in Results and Discussion: past and present tense

- In the Results and Discussion sections, the past tense is generally used to summarize the findings.
- But when you are interpreting the results or describing the significance of the findings, the present tense should be used.
- Also use the present tense to refer to tables, figures, and graphs that you are using to present your results, for example: *Table 4 shows the blood pressure levels of patients before and after administration of the recommended dose of the drug.*
- Often, a **combination** of both the past and the present tense is used in sentences within the discussion section.

Example: *Two thirds of the children **demonstrated** an elevated level of at least one risk factor, indicating that children with obesity **are** at an increased risk of cardiovascular diseases.*

- Note that the first part of the sentence refers to the results; hence the past tense has been used for this part. On the other hand, the present tense has been used for the second part as this part explains what the result means.



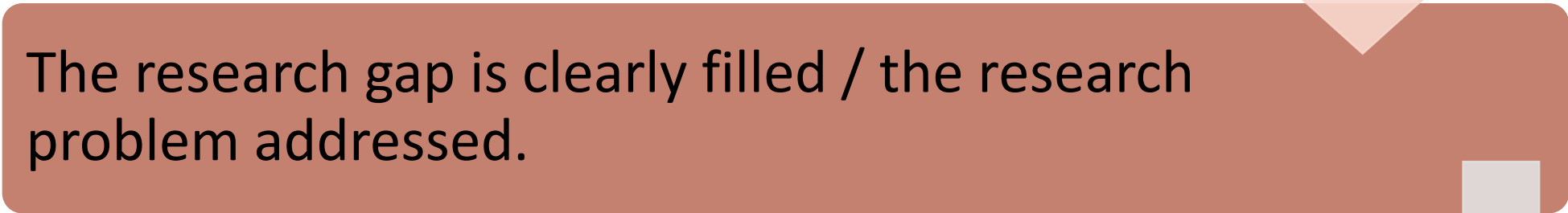
(Busse & August 2020)

Discussion contents

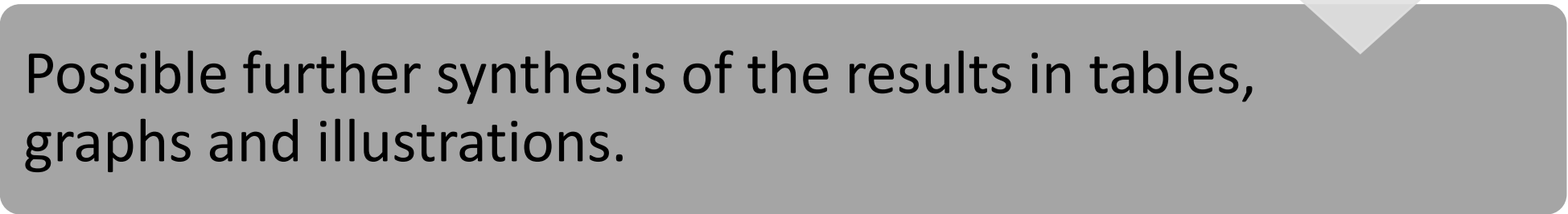
The results of the study are interpreted and put in context. Justification is given for the interpretation.

A large orange rounded rectangle containing the first point. A light orange arrow points downwards from its bottom right corner towards the second box.

The research gap is clearly filled / the research problem addressed.

A large brown rounded rectangle containing the second point. A light brown arrow points downwards from its bottom right corner towards the third box.

Possible further synthesis of the results in tables, graphs and illustrations.

A large grey rounded rectangle containing the third point. It is the final box in the sequence.

Conclusion contents

(Incorporated into the Discussion or under separate heading)



Reiterates (repeats) the research question and its answer



States the significance of that answer



Identifies and discusses directions for future research and practice



Outlines limitations and issues that were beyond the scope of the study. (The limitations are often stated under a separate heading.)



Some elements of Discussion and Conclusion sections

- References to literature in the field to contextualise research
- References to the study aims, hypothesis or research question
- Restating the result or some pertinent results
- Indicating the results that confirm or contradict previous findings
- Offering explanations of the findings
- Advising cautious interpretation of the findings, using hedging language (see below)
- Commenting on and noting implications of the findings (could be part of Conclusion)
- Making suggestions for future research (often in Conclusion section)
- State limitations (often in separate section)

Moves in the Discussion and Conclusion sections (reminder)



1. **Comment on key results** (making claims, explaining the results, comparing the new work with the previous studies, offering alternative explanations). Summarize the key findings and **contextualise** them against the background of other studies.



2. State the **limitations** of the study (Could be part of a separate section on limitations)



3. Make recommendations for future implementation and/or for future research. This move contains the **final message sentence**, stating the **main findings and implications** in one sentence.

Let us walk through the Discussion and Conclusion section together (*cf.* Busse and August 2020):

Step by step through the Discussion and Conclusion section (1)

1. Begin with a brief **restatement of the main findings** in a few carefully crafted sentences.
2. **Interpret the significance** of these results, focusing on **general applications** of the study's specific findings too.
3. **Compare these study findings to other research.** Do they agree or disagree with those from other studies? Does this study shed further light on accepted theories? Explain your findings against the background of scientific literature, focusing on pathways or mechanisms that might be linked to or cause the results.

Referring to own research

Metadiscourse is common (talking about your study), e.g.

This study set out to ...

This paper argued that ...

This essay has discussed the reasons for ...

In this investigation, the aim was to assess

Sometimes **'we'**:

- We assessed...
- We aimed to determine

'Research' is a mass noun that does NOT have a plural form, so ~~researches~~ is wrong. If you want to use a plural form, you can write 'research studies/projects/investigations'

A **'work'** in English is normally a very long text, longer than an article. Do not refer to your paper/article as 'this work'.

Example of a summary at the beginning of a Discussion section

The primary aim of this study was to assess the impact of cell phone use on crash risk in a naturalistic driving setting. It was determined that cell phone use increased the odds of a culpable crash by 70%. This association was persistent across certain subgroups: male drivers, drivers aged between 26 and 65 years, drivers with a full license, drivers not suspected of using alcohol/drugs and for injury and non-injury crashes. These findings lend support to the limited body of epidemiological evidence showing that cell phone use while driving increases crash risk.^{21,23,24}

Mark Asbridge, Jeff R Brubacher, Herbert Chan, Cell phone use and traffic crash risk: a culpability analysis, *International Journal of Epidemiology*, Volume 42, Issue 1, February 2013, Pages 259–267, <https://doi.org/10.1093/ije/dys180>

Restate the purpose of the study

Summarize the major finding

Contextualize by referring to other research

Discussion: first part example sentences

- **We show that** substituting 50% of ASF with novel alternatives can lead to profound system-wide impacts. Unlike previous studies that assessed dietary changes with novel foods, in this study we considered a more realistic composition of the plant ingredients that would be used to produce novel alternatives and analyzed them in a dynamic system-wide global framework.
- Increasing the substitution rate beyond 50% led to only limited additional reductions of deforestation and associated emissions because substituting 50% of AFS with novel alternatives effectively eliminates land use change driven by those commodities. Yet, the abandonment of previously productive agricultural land offers the possibility of additional measures to restore land and sequester carbon, allowing for more than doubling of GHG emission reductions.

(Kozicka, M., Havlík, P., Valin, H. *et al.* Feeding climate and biodiversity goals with novel plant-based meat and milk alternatives. *Nat Commun* **14**, 5316 (2023). <https://doi.org/10.1038/s41467-023-40899-2>)

Analyse the first part of the of the **Discussion and Conclusion section(s)** in one or more representative research article(s) from your field, or in the given examples on this or previous slides

1. Does the author restate the study objective, the research question or hypothesis?
2. Does the author compare the results to the research question / hypothesis, and summarize how the study fulfils the gap stated in the introduction?
3. Is supporting evidence provided? In what form?
4. Is a new mechanism or model described?

[AIDS](#) (Industrial mining and HIV risk: evidence from 39 mine openings across 16 countries in sub-Saharan Africa)

[Genomic Evolution of Breast Cancer Metastasis and Relapse - ScienceDirect](#)

Move 2 | Context

- Use language to **compare and contrast** to agree with previous research (cite supporting studies) or to point out discrepancies (cite contradicting studies)
- Use **linguistic hedges** to weaken the claims that are made

Likewise,... / Similarly... / In line with previous findings... / In agreement with...

In contrast.../ Unlike... / However,... / Whereas...

Adverbs that introduce uncertainty:
possibly, potentially

Modal verbs:
can/could, may/might

Example of contextualizing a study in the context of other research (2)

“Our results, consistent with a number of studies of other species (3, 4) suggest that body mass, rather than CIs (condition indices), may be one of the most useful measures for linking nutritional changes to population dynamics.”..

“Numerous studies have documented higher growth rates in mammals with increased nutritional intake [85, 86, Table 1 in 87] and corresponding higher juvenile survival [31, 46, 85, 88], reproductive success [89, 90], and survival rates of larger individuals [91] which suggests there is a benefit to a younger animal investing in structural size...

Similarly, in our study, cub survival was associated with measures that reflect overall size (i.e. length and body mass) rather than energy reserves alone.”

Identifying reliable indicators of fitness in polar bears | PLOS ONE

Step by step through the Discussion and Conclusion section(2)

Continue through the Discussion and Conclusion section (cf. Busse and August 2020):

4. **Candidly report and discuss the strengths and limitations of your study** to help the readers interpret your research findings correctly. Some journals require separate section headings, while others prefer having these limitations embedded in paragraphs.
5. **Summarise the potential impact and applications** of your research study. How could the results be used to design an intervention? What implications might the study have for clinical protocols or public policy?
6. **Conclude the discussion with specific suggestions for future work**. Illuminate specific remaining gaps in the literature that necessitate further study. Instead of stating too generally that 'further research is necessary', rather provide specific and practical recommendations for further studies.

Example sentences: limitations, impact and suggestions

- Although this study did not model the market for novel alternatives, there are many possible future scenarios of market development and novel alternatives adoption. We argue that the 50% substitution scenario is a realistic one....

Common discussion section pitfalls and recommendations

(Busse and August

Pitfall	Recommendation
The author repeats detailed results or presents new results in the discussion section.	Focus on how your study fits into the scientific literature. Specific values should not be repeated in the discussion. It is also not appropriate to include new results in the discussion section.
The author fails to describe the implication of the study's limitations.	Where possible, identify the implications of your study's limitations. No matter how well-conducted and thoughtful, all studies have limitations. Candidly describe how the limitations affect the application of the findings.
Statements about future research are too generic.	Be specific! Is the relationship between exposure and outcome not well-described in a population that is severely impacted? Or might there be another variable that modifies the relationship between exposure and outcome? This is your opportunity to suggest areas requiring further study in your field, steering scientific inquiry toward the most meaningful questions.

Essential tips for the Discussion/Conclusion section (Day and Gastel, 2011:73):

- Present the principles, relationships and generalisations shown by the results. Don't repeat the results: refer to them as necessary and discuss them.
- Point out exceptions or lack of correlation. Do not be dishonest or cover up data that does not fit.
- Show how your results and interpretations agree or contrast with previously published studies.
- Discuss the theoretical implications and practical applications of your work confidently enough, but cautiously.
- State your conclusions as clearly as possible.
- Summarize your evidence for each conclusion.



Limitations of the study

- Journals vary in their approach to strengths and limitations sections: some are embedded paragraphs within the discussion section, while some mandate separate section headings.
- Keep in mind that every study has strengths and limitations.
- Candidly reporting yours helps readers to correctly interpret your research findings.
- If you don't point out the limitations, your reviewers will!



Task: Steps in the Discussion and Conclusion

Examine some articles in your field:

Which of the 6 steps can you identify? How are they expressed? Compare your findings to your colleagues’.

1. Restatement of the main findings
2. Interpretation of their significance, and applications
3. Comparison to other research
4. Discussion of strengths and limitations of the study
5. Summary of the potential impact and applications of the study
6. Specific suggestions for future work



Data commentary: verbal description of visualised data

Includes certain elements, in the following order, the first 2 being in Results, the 3rd in **Discussion/Conclusion**.

1. **Location** elements/statements referring to visuals , followed by **summary statements**
2. **Highlighting** statements
3. **Discussions of implications, problems, exceptions, recommendations, or other interesting and relevant aspects of the data , including making claims based on the data.**

Referring to own research (examples in script)



Metadiscourse is common (talking about the study), e.g.

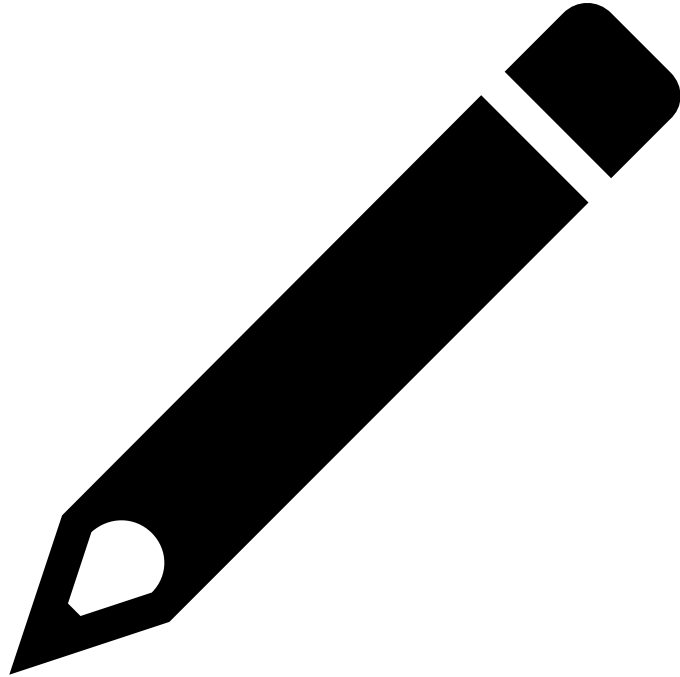
- This study set out to ...
- This paper argued that ...
- This essay has discussed the reasons for ...
- In this investigation, the aim was to assess

Also, '**we**':

- We assessed...
- We aimed to determine

'Research' is a mass noun that does NOT have a plural form, so ~~researches~~ is wrong. If you want to use a plural form, you can write 'research studies/projects/investigations'

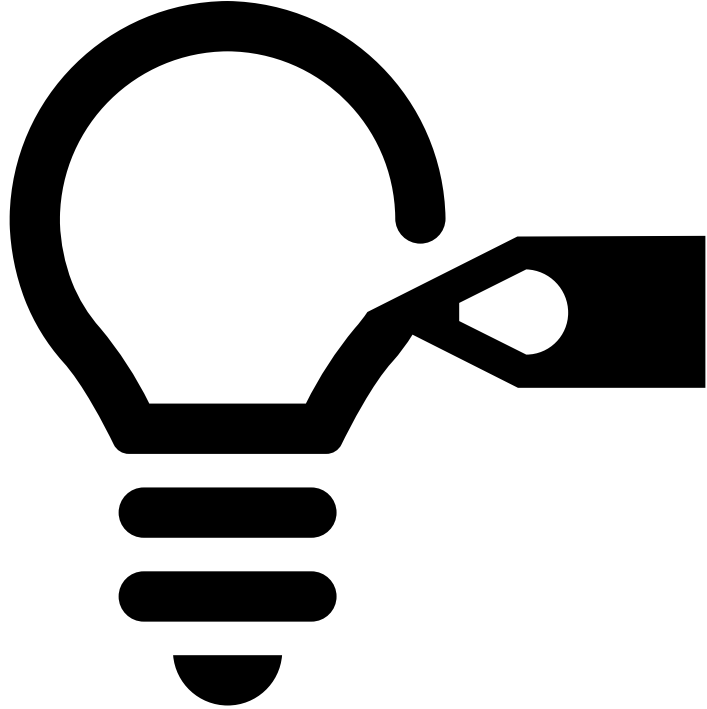
A '**work**' in English is normally a very long text, longer than an article. Do not refer to your paper/article as 'this work'.



Task: Language use in these sections 1

Examine some articles/thesis in your field:

1. Identify at least 3 **claims** made about the results.
2. How is the **strength or weakness** of claims expressed? Write down 2 ways.
3. Highlight the verbs that are used. What **verb uses dominate** in this section? List at least 5 verbs here.
4. What **degree of certainty** is expressed by the verbs?
5. Do the authors **distance** themselves from the claims they make? How?



Task: Language use in these sections 2

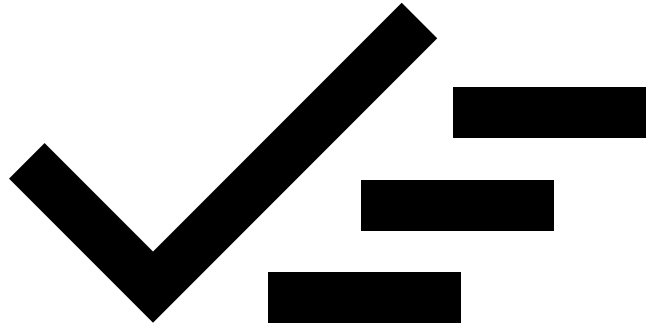
6. Are the claims **linked** to **previous research**? If so, how?
7. How are the **comparisons** made?
8. Do the authors make **generalizations**? Verb use? Grounded in reasonable evidence?
9. Do they indicate any **exceptions** to the claims/generalizations that they made? How?
10. What **general trends** in the language use emerge? Compare your findings with those of your colleagues.

Hedging: cautiously expressing claims based on research

Be confidently uncertain!

Hedging = qualifying, softening or moderating a claim in various ways

Ensure the strength of hedging is appropriate for the claim.



Hedging techniques 1

(See script for examples)

Likelihood:

- Modal verbs (can, could, may, might, etc.)
- Phrases to indicate strength (clearly, likely, slightly, etc.)
- Adjectives (e.g., 'little' as in There is **little** possibility that different weather conditions affected the noise sources. Note: not 'a little')

Distance:

- Leave room for doubt in the claim.
- Attribute the point to someone else/link to other studies, e.g. *According to some earlier studies (citation)/In view of several scholars (citation)...*

Hedging techniques 2

(See script for examples)

Soften generalizations

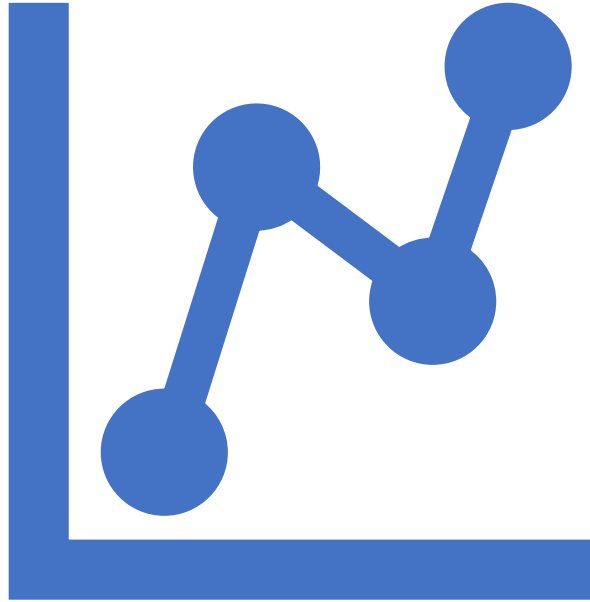
- Verbs such as appear to, tend to, seem to
- Qualify/limit the subject (many, a majority of, some, in most..)
- Add exceptions (With the exception of, apart from, except for)

Use weaker verbs

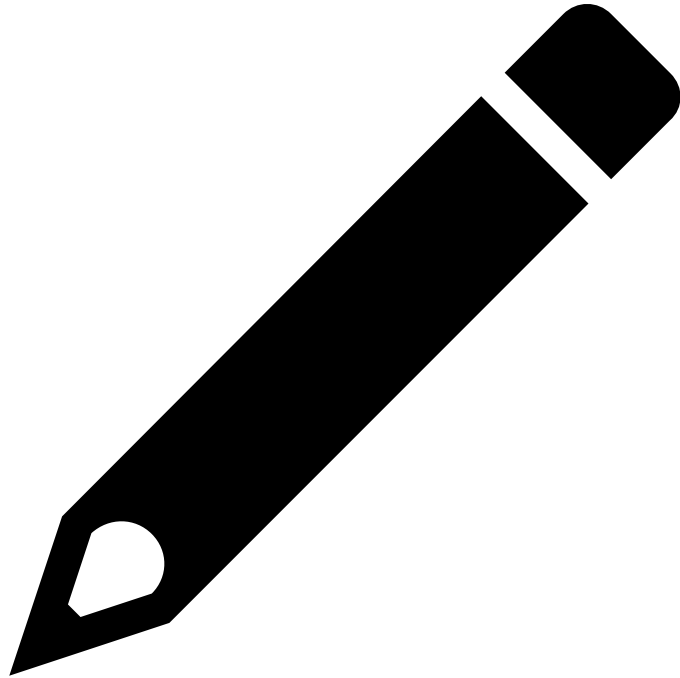
Such as *estimate, suggest, indicate, appear to, seem to, tend to, for example*
use 'contribute to' rather than 'cause', 'suggest' rather than 'prove'

Refer to the *Academic Phrasebank* for many examples of hedging phrases to use.

Generalisations – typical in Discussion



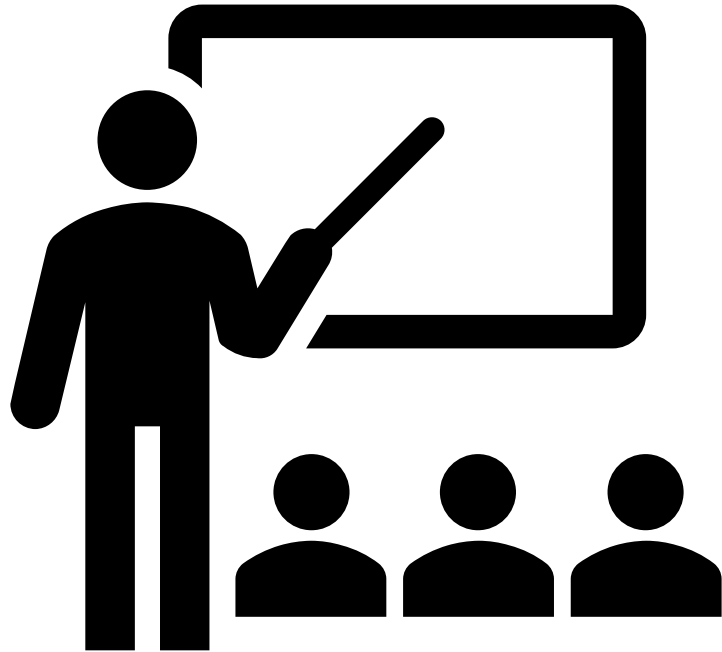
- Make general claims based on specific findings in the study
- Supported by reasonable evidence and stated cautiously
- Justified generalisations: based on robust results
- Unjustified generalisations make claims that are not based on data or that are too strong
- Overgeneralisations should be avoided in academic writing: e.g.
 - ❑ All Americans have bad eating habits.



Exercise: soften generalizations Adapted from (Swales & Feak 2012:164)

1. Tall people have higher incomes than short people.
2. Thin people get better jobs than fat people.
3. People with foreign-sounding surnames can't find jobs.
4. Economic sanctions are ineffective.
5. Alcohol causes brain damage.
6. Passive smoking causes cancer.
7. Recycling is the best solution to the waste disposal problem.
8. Physical exercise prevents and relieves depression.

Exercise: Choose the verb that makes the weaker claim



1. The results *indicate/establish* that there is a link between smoking and lung cancer.
2. The survey results *suggest/prove* that ...
3. The latest series of studies *challenge/question* the value of...
4. Changes in ambient temperature may have *influenced/distorted* the results.
5. The results *indicate/establish* that there is a link between smoking and lung cancer.

Analyse the Discussion and Conclusion section(s) in one or more representative research article(s) from your field

1. How does the author contextualize the results of this study within the context of other studies?
2. Does the author use hedging language when making claims based on the results? Identify examples of hedging devices, for example:
 - Modal verbs
 - Qualifying adjectives
 - Using weaker verbs
 - Distancing the author from claims



5-10 minutes



Making comparisons

Frequent in Discussion sections

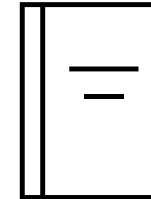
- **Few' and 'many' are used with countable nouns**, such as 'few/many studies'. The comparative form of 'few' is 'fewer' and the superlative form is 'fewest'. The comparative form of 'many' is 'more' and the superlative form is 'the most'.
- **'Little' and 'much' are used with uncountable nouns**, such as 'little/much information'. The comparative form of 'little' is 'less' and the superlative form is 'the least'. The comparative form of 'much' is 'more' and the superlative form is 'the most'.
- 'Little' and 'a little' are not the same. 'Little' means 'almost nothing'. 'A little' means some.

Comparing and contrasting : Academic Phrasebank

(<https://www.phrasebank.manchester.ac.uk/compare-and-contrast/>).

Many useful stock phrases to

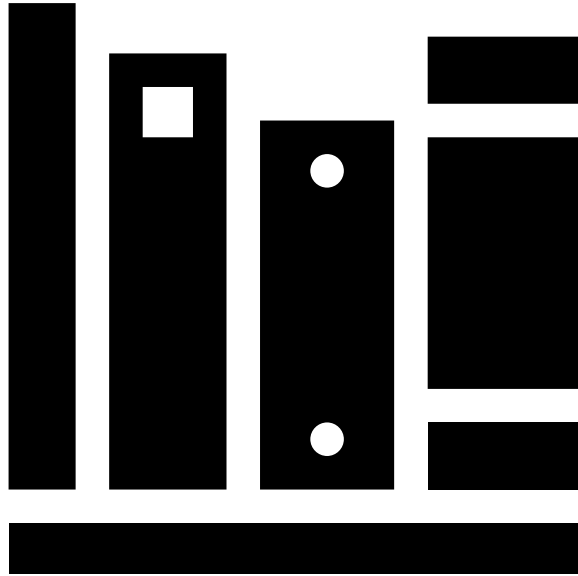
- Introduce similarities and differences
- Compare within sentences (comparative forms)
- Indicate similarities and differences across two sentences



Examples:

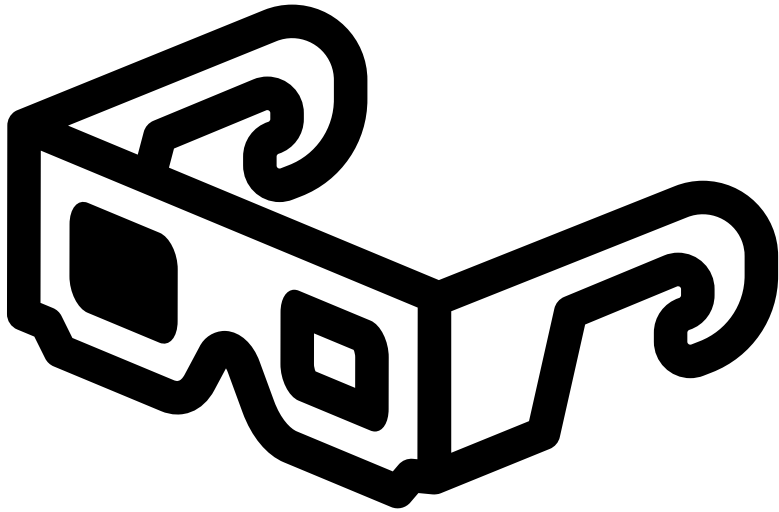
This interpretation differs from that of Smith and Jones (2004) who argue that...

Test X was more /less reliable than test Y at identifying...



Linking expressions to compare and contrast

- On the one hand... on the other hand
- Similarly, likewise
- Conversely
- Unlike
- Whereas, while
- Rather than
- The reverse is true
- There is a contrast between



More language use for comparisons: correct prepositions

- A comparison **between** (two different things)
- A comparison **of** (different examples of the same thing)
- In comparison **to/with**
- As compared **to**
- Analogies **between**
- Differences and similarities **between/in**
- The distinction **between**
- Different **from**
- Similar **to**

The Conclusion section



Short section



Sometimes optional if included in Discussion section.



Looks backward by summarising the main area(s) covered in the paper and commenting on it/them



Looks forward by including suggestions for improvement and making recommendations for future work

Content of Conclusion section



- Sometimes restating the aims or research question of the study
- Summarising main research findings
- Stating implications for the field of study
- Explaining the significance or contribution of the study
- Recognising the limitations of the study (If not in separate section)
- Making recommendations for future research work, practice or policy

Move 3 | Limitations

- **Make concessions** to admit difficulties in interpretation or provide alternative explanations of your findings.

Although X is true, Y also appears to be true.

- Expressions to **state limitations**:

We would like to point out that... / It should be noted that... / Our study is restricted to / only looked at / only addressed...

- Expressions to **explain unexpected data**:

*These unexpected results **may** be due to... / This discrepancy **may** be attributed to... / This finding **can be accounted for** by... / The lack of significance is **probably** a consequence of...*

- Ensure that you use appropriate **hedges**

Stating limitations and exceptions: examples

1. It should be noted that this study has been primarily concerned with...
2. This analysis has concentrated on...
3. The findings of this study are restricted to...
4. This study has addressed only the question of...
5. The limitations of this study are clear: ...
6. We would like to point out that we have not...
7. However, the findings do not imply...
8. The results of this study cannot be taken as evidence for...
9. Unfortunately, we are unable to determine from this data...
10. The lack of... means that we cannot be certain...
11. Notwithstanding its limitations, this study does suggest...
12. Despite its preliminary character, the research reported here would seem to indicate...
13. However exploratory, this study may offer some insight into...

See Academic Phrasebank for more examples.



Final comment: You don't get a second chance to make a final good impression!



END CONFIDENTLY AND
POSITIVELY



BE FACTUAL AND NOT
BOASTFUL

Analyse the overall structure of the **Discussion and Conclusion section(s)** in one or more representative research article(s) from your field, or in one of the articles provided

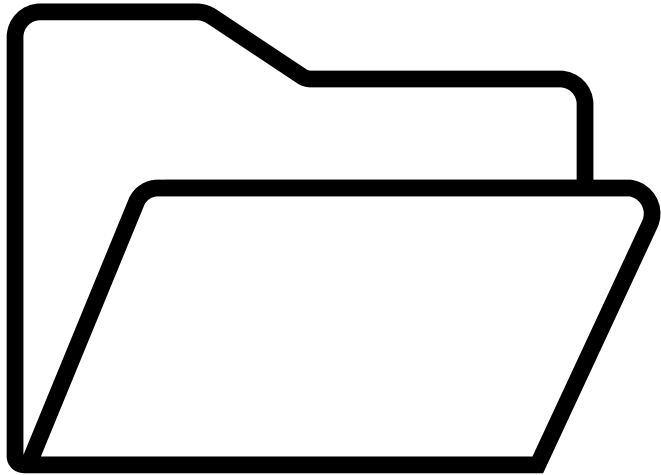
1. Focus on the heading(s) of the section(s): Are there separate Discussion and Conclusion sections, or are they combined into one?
2. What is included in the Discussion section? Can you identify the following moves?
 - Summary of findings / take-home message
 - Context
 - Limitations
 - Impact and recommendations for future research
3. How do the Results and Discussion sections differ?
4. Analyse the use of tenses in the Discussion.

AIDS (Industrial mining and HIV risk: evidence from 39 mine openings across 16 countries in sub-Saharan Africa)

[Genomic Evolution of Breast Cancer Metastasis and Relapse - ScienceDirect](#)

Assignment 6A – see instructions document

- Write a 250-word Discussion & Conclusion section for your study/proposal. Students who do not have results yet, consider/envisage what results they hope to discuss, and describe them in a Discussion/Conclusion section (as for a proposal), or else as a section of a research article.
- In the Interpretation/ Discussion section, interpret your results, while also giving justification for your interpretation and putting the results in the context of wider research. You could further synthesize the results in tables, graphs and illustrations.
- In the Conclusion section, discuss the implications of the results of the study for future research and practice. In addition, outline limitations, and issues beyond the scope of the study.
- Follow the six steps of Discussion and Conclusion (Busse and August, 2020) outlined on the slides and in the script.
- Highlight the main results, justification, implications and issues beyond the scope of the study, as well as the use of hedging techniques and devices to increase cohesion in coherence in the text



Portfolio

Remember that, in order to earn the 2 credits for the course, you need to submit a portfolio consisting of:

1. ONE file in WORD format, containing
 2. the **first, unreviewed version** of each assignment
 3. followed by the **final improved version**
 4. of each of the **FOUR reviewed assignments**.
- The portfolio text should contain **no comments or markup**, apart from indicating the **moves or sections**.
 - Focus on improving the texts according to my comments and those of your reviewers.
 - Name the portfolio surname.name Portfolio

Language problems in assignments

- Download the English grammar and spelling software that goes with WORD and revise your writing with that before submission. Set your proofing language to English!! Revise by uploading your texts to TRINKA!
- Overuse/Incorrect use of the present continuous tense. Only use it to describe current trends, for example 'Scientists are currently considering new treatment options.', or actions taking place at the moment of speaking.
- Otherwise, use the simple present tense: 'Post-traumatic Stress Disorder (PTSD) is a mental health disorder which occurs after experiencing one or more traumatic events.'

Language work: See script for exercises and examples

Defining Relative clauses

- The information defines the noun and is essential to the meaning of the sentence.
- NO COMMAS are used with defining relative clauses.

Example: Leonardo Da Vinci was the first person *who designed a helicopter-like flying machine*.

- Pronouns *that, who, whose, which, whom*

There is no comma before '*that*' in English.

Non-defining relative clauses

(See script for exercises and examples)

- The information added might be useful but the sentence would still be meaningful without it.
- Commas separate the clause from the sentence.

Example: Leonardo Da Vinci, *whose designs were spectacularly ahead of his time*, was famous as a painter, sculptor, architect, musician, mathematician, engineer, botanist and writer.

Use of commas

(See script for examples)

- Phrases beginning with ‘including’, ‘according’, etc. are preceded by a comma and sometimes bracketed off with commas.
- Separate items in a series or list, but they are often not used between the last two items
- Separate non-defining relative clauses from the main part of the sentence
- Separate ‘sentence adverbs’ from the rest of the sentence

Do not use commas in these cases (comma splice mistake)

Do not use a comma to link two sentences together. Rather use a full stop or a semi-colon, or another linking device.

- ~~All contributions are welcomed, no donation is too small.~~
- ~~The first experiment was unsuccessful, on the other hand, the second one soon yielded the expected results.~~

Corrected examples:

- All contributions are welcomed; no donation is too small.
- All contributions are welcomed. No donation is too small.
- The first experiment was unsuccessful; on the other hand, the second one soon yielded the expected results.
- The first experiment was unsuccessful. On the other hand, the second one soon yielded the expected results.



Thank you
for your attention.